

ARVENA

Master Product Requirements Document — v1.0

Status: FINAL PRODUCT BLUEPRINT

Tanggal: 7 September 2026

Nama Produk: ARVENA

Tagline: Everything connected. Built to grow.

1. EXECUTIVE SUMMARY

ARVENA adalah **Service Business Operating System** untuk bisnis jasa yang memiliki pelanggan, pekerjaan, jadwal, tim lapangan, tagihan, follow-up, dan pekerjaan berulang.

ARVENA bukan hanya CRM.

ARVENA bukan hanya aplikasi teknisi.

ARVENA bukan hanya sistem WhatsApp.

ARVENA adalah **pusat operasional bisnis jasa**.

Alur utama:

Customer → Lead/Request → Quote → Job → Visit → Worker → Completion → Invoice → Payment → Follow-up → Repeat/Recurring

ARVENA harus membuat owner dapat mengetahui:

- apa yang perlu diperhatikan sekarang;
- pekerjaan apa yang sedang berjalan;
- siapa yang mengerjakan;
- customer mana yang belum di-follow-up;
- invoice mana yang belum dibayar;
- pekerjaan mana yang terlambat;
- customer mana yang berpotensi repeat order;
- bagaimana performa bisnis dan tim.

Prinsip utama:

One Input → Many Outcomes

Contoh:

Admin membuat satu Job.

Dari Job tersebut ARVENA otomatis menghasilkan:

- jadwal;
- assignment worker;
- tampilan job di HP worker;
- customer history;
- dashboard owner;
- activity timeline;
- reminder;
- dasar invoice;
- dasar laporan;
- follow-up setelah pekerjaan selesai.

Data tidak boleh dimasukkan berulang kali.

2. BRAND FOUNDATION

2.1 Nama

ARVENA

Makna brand:

Sistem yang menghubungkan seluruh bagian bisnis agar tumbuh dengan rapi dan terarah.

2.2 Filosofi Logo

Logo resmi ARVENA menggunakan:

- bentuk **A** sebagai identitas Arvena;
- akar sebagai fondasi;
- tunas/daun sebagai pertumbuhan.

Makna:

Akar = Customer + Data

Batang = ARVENA sebagai pusat operasional

Cabang = Sales, Jobs, Workers, Schedule, Finance, Automation

Daun = Pertumbuhan bisnis

2.3 Brand Personality

ARVENA harus terasa:

- profesional;
- tenang;
- modern;
- mudah digunakan;
- dapat dipercaya;
- tidak terlalu korporat;
- tidak terlalu playful;
- premium tetapi tetap cocok untuk UMKM.

2.4 Brand Colors

Baseline dari logo resmi:

Arvena Forest #073525

Digunakan untuk:

- logo;
- CTA utama;
- sidebar aktif;
- primary navigation;
- key metrics.

Arvena Sage #698D70

Digunakan untuk:

- secondary actions;
- success indicators;
- charts;
- supporting visuals.

Arvena Soft Sage #98ACA4

Digunakan untuk:

- borders;
- background accents;
- disabled/supporting UI.

Arvena Canvas #FDFDFB

Digunakan sebagai background utama.

UI tidak boleh penuh hijau.

Hijau digunakan sebagai identitas dan hierarchy, bukan dekorasi berlebihan.

3. PRODUCT VISION

ARVENA ingin menjadi:

Operating system utama untuk bisnis jasa dari usaha kecil hingga perusahaan multi-cabang.

Perjalanan target:

Indonesia → Asia Tenggara → Asia → Global

Karena itu data model tidak boleh dibuat khusus Indonesia.

ARVENA dari awal harus mendukung:

- country;
 - timezone;
 - currency;
 - locale;
 - international phone format;
 - configurable taxes;
 - regional date format;
 - configurable address structures.
-

4. CORE PRODUCT PROMISE

ARVENA harus membantu bisnis:

1. Tidak kehilangan order

Lead dan request harus tercatat.

2. Tidak lupa follow-up

ARVENA mengingatkan.

3. Tidak kehilangan kontrol pekerjaan

Owner tahu status setiap pekerjaan.

4. Mengurangi pekerjaan admin

Data tidak diketik berulang.

5. Mengontrol tim lapangan

Owner mengetahui assignment dan progress.

6. Menagih lebih cepat

Job dapat menghasilkan invoice.

7. Mendapat repeat order

Customer history dan recurring service digunakan kembali.

8. Menumbuhkan bisnis

Owner mengetahui masalah dan peluang melalui Business Health Center.

5. PRIMARY PRODUCT PRINCIPLES

P1 — One Input, Many Outcomes

Data utama dimasukkan sekali.

Sistem menyebarkan hasilnya ke modul terkait.

P2 — Job ≠ Visit

Ini merupakan keputusan arsitektur permanen.

Job = pekerjaan/customer request secara keseluruhan.

Visit = kunjungan aktual ke customer.

Contoh:

Job:

Instalasi CCTV Gudang ABC

Visits:

1. Survey
2. Instalasi
3. Testing
4. Training customer

Satu Job dapat memiliki satu atau banyak Visit.

P3 — WhatsApp Is a Connector

ARVENA tidak boleh bergantung pada WhatsApp.

Jika WhatsApp bermasalah:

- customers tetap bisa digunakan;
 - jobs tetap berjalan;
 - schedule tetap berjalan;
 - invoice tetap berjalan;
 - follow-up manual tetap berjalan.
-

P4 — Role-Based Simplicity

Setiap user hanya melihat hal yang relevan.

Worker tidak membutuhkan dashboard revenue.

Finance tidak membutuhkan pipeline sales.

Sales tidak membutuhkan system settings.

P5 — Exception-Driven UX

Owner tidak membutuhkan 20 grafik.

Hal pertama yang harus dilihat:

Perlu perhatian

Contoh:

- 7 leads belum dihubungi
 - 3 jobs belum memiliki worker
 - 2 worker terlambat
 - 4 invoice overdue
 - 8 follow-up harus dilakukan hari ini
-

P6 — Mobile First for Field Work

Worker dapat bekerja penuh dari smartphone.

Owner juga harus dapat melakukan operasi penting lewat smartphone.

P7 — Global-Ready

Tidak ada hardcoded:

- Rp;
 - +62;
 - Jakarta timezone;
 - province Indonesia;
 - format alamat Indonesia.
-

P8 — Cost-Aware SaaS

ARVENA tidak memberikan unlimited untuk resource yang menghasilkan biaya variabel.

Contoh:

- WhatsApp messaging;
 - AI;
 - storage;
 - automation volume;
 - API calls.
-

6. TARGET BUSINESSES

ARVENA menggunakan:

Universal Core + Industry Templates + Optional Modules

Satu codebase.

Bukan aplikasi berbeda untuk setiap industri.

7. INITIAL INDUSTRY TEMPLATES

Template awal antara lain:

Home & Property Services

- AC/HVAC
- CCTV
- electrical
- plumbing
- pump
- water heater
- genset
- solar
- handyman
- renovation
- painting
- roofing
- waterproofing

Cleaning

- home cleaning
- office cleaning
- sofa cleaning
- carpet cleaning
- mattress cleaning
- window cleaning
- pressure washing

Pest & Outdoor

- pest control
- termite
- fogging

- gardening
- landscaping
- pool maintenance
- tree service

Technology

- WiFi installation
- ISP
- fiber
- CCTV
- network
- IT support
- computer repair
- printer repair
- server maintenance

Automotive

- workshop
- mobile mechanic
- detailing
- car wash
- tire
- battery
- towing
- car AC

Equipment Services

- industrial machines
- compressors
- generators
- forklifts
- cold storage
- refrigeration
- maintenance contracts

Personal/Home Services

- laundry
- pet grooming
- barber/home beauty
- moving
- waste services
- water tank cleaning

Event Services

- photography
- videography
- decoration
- sound system
- lighting
- equipment rental
- catering

Industry template menentukan:

- terminology;
- default statuses;
- checklist;
- suggested fields;
- recommended modules;
- default workflow.

Industry template **tidak membuat codebase baru.**

8. OPTIONAL MODULE ENGINE

ARVENA Core dapat ditambah modul.

Initial Modules

Assets

Untuk:

- AC
- CCTV
- machinery
- vehicles
- network hardware

Recurring Service

Untuk:

- cleaning
- pest control
- maintenance
- pool
- subscription service

Subscription

Untuk:

- ISP
- managed maintenance
- recurring service billing

Inventory

Untuk:

- workshops
- equipment repair
- CCTV installers

Project

Untuk:

- renovation
- event projects
- installation projects

Ticket & SLA

Untuk:

- IT support
- ISP
- B2B maintenance

Rental

Untuk:

- equipment
 - event supplies
 - vehicle/equipment rental
-

9. USER ROLES

Owner

Can:

- access all organization data;
 - dashboard;
 - customers;
 - leads;
 - jobs;
 - visits;
 - workers;
 - finance;
 - reports;
 - settings;
 - integrations;
 - billing.
-

Admin / Dispatcher

Can:

- customers;
- jobs;
- schedule;
- visits;
- assignments;
- day-to-day operations.

Financial access configurable.

Sales

Can:

- leads;
 - customers;
 - quotes;
 - follow-up;
 - assigned opportunities.
-

Worker / Technician

Can only see:

- assigned jobs;
 - assigned visits;
 - customer location;
 - checklist;
 - work notes;
 - photos;
 - job status.
-

Supervisor

Can:

- assigned team;
 - jobs;
 - workers;
 - operational reports.
-

Finance

Can:

- invoices;
 - payments;
 - receivables;
 - finance reports.
-

10. PERMISSION MODEL

Every organization has independent access control.

Base permissions:

view create update delete assign approve export manage

Ultra supports custom roles.

Critical records cannot be permanently deleted by normal users.

Use:

Archive / Soft Delete

Hard deletion requires privileged actions.

11. ORGANIZATION MODEL

Every business exists as:

organization

Every business record must contain:

organization_id

Examples:

- customers
- leads
- jobs
- visits
- invoices
- workers
- notes
- assets
- activities

No record may exist without tenant ownership unless explicitly global/configuration data.

12. ONBOARDING

Goal:

First useful job in under 10 minutes.

Flow:

Step 1

Create account.

Step 2

Business information:

- business name;
- industry;
- country.

Step 3

System automatically determines defaults:

- currency;
- timezone;
- locale;
- tax behavior.

User may edit them.

Step 4

Select business template.

Step 5

Add first team member.

Optional skip.

Step 6

Add first customer.

Options:

- manually;
- CSV import.

Step 7

Create first Job.

Onboarding complete.

WhatsApp connection:

Optional

It must never block onboarding.

13. MAIN NAVIGATION

Desktop sidebar:

Home

Dashboard

Sales

Leads

Quotes

Follow-ups

Operations

Jobs

Calendar

Customers

Assets

Finance

Invoices

Payments

Team

Team

Performance

Insights

Reports

System

Integrations

Settings

Modules not enabled must not clutter navigation.

14. MOBILE NAVIGATION

Owner mobile bottom navigation:

Home | Jobs | Customers | Calendar | More

Worker mobile:

Today | Jobs | Notifications | Profile

Worker interface intentionally has fewer modules.

15. OWNER DASHBOARD

Dashboard must answer:

What needs my attention today?

Top section:

Perlu Perhatian

Cards:

- unassigned jobs;
- overdue jobs;
- late visits;
- unpaid invoices;
- follow-ups due;
- schedule conflicts.

Each card must be actionable.

Example:

4 Invoice Overdue — Rp8.400.000

Click opens filtered invoice list.

Today's Operations

Show:

- today's jobs;
 - visits;
 - workers;
 - completion status.
-

Money Snapshot

Show:

- issued invoices;
 - paid;
 - outstanding;
 - overdue.
-

Team Snapshot

Show:

- workers active today;
 - jobs completed;
 - jobs delayed.
-

Business Health Center

ARVENA generates recommendations.

Examples:

12 leads belum di-follow-up.

4 invoice overdue lebih dari 14 hari.

27 customer tidak melakukan transaksi selama 6 bulan.

6 recurring services harus dijadwalkan minggu ini.

Dashboard should prioritize actions over decorative charts.

16. CUSTOMER MANAGEMENT

Customer types:

- Person
- Company

Fields:

- name;
- normalized phone;
- WhatsApp phone;
- alternate phone;
- email;
- customer type;
- address;
- location coordinates;
- tags;
- source;
- notes;
- created date;
- last activity.

17. PHONE NORMALIZATION

Phone must be normalized into canonical international format.

Example:

08123456789

becomes:

+628123456789

Deduplication happens:

per organization

Not globally.

One phone may exist in separate businesses.

18. CUSTOMER DETAIL

Customer page includes:

Overview

- contact;
- address;
- tags;
- owner/sales.

Jobs

Historical jobs.

Assets

Customer equipment.

Quotes

Quote history.

Invoices

Billing history.

Payments

Payment history.

Notes

Internal notes.

Activity

Timeline.

Follow-ups

Upcoming actions.

19. LEAD MANAGEMENT

Customer and Lead are separate entities.

Customer:

Person/company.

Lead:

Opportunity/request.

One Customer may have multiple Leads.

Example:

Customer: PT ABC

Lead 1: Install CCTV

Lead 2: Add access control

20. LEAD PIPELINE

Default statuses:

New → Contacted → Qualified → Quote → Negotiation → Won / Lost

Industry templates may change labels.

Example AC:

New Request → Scheduled → Converted

Pipeline must be configurable.

21. LEAD PAGE

Desktop:

Table / Pipeline toggle.

Fields:

- customer;
- phone;
- service/product;
- status;
- temperature;
- owner/sales;
- source;
- next follow-up;
- created;
- updated.

Mobile:

Cards.

Actions:

- View
- WhatsApp
- Call
- Edit
- Convert
- Follow-up

22. LEAD FILTERS

Support:

- search customer;
- phone;
- status;
- sales;
- source;
- temperature;
- overdue follow-up;
- date range.

Saved views may be added later.

23. QUOTATION

Quote is optional.

Job may be created without a Quote.

Quote fields:

- customer;
- items;
- quantity;
- unit;
- price;
- discount;
- tax;
- notes;
- expiry;
- terms.

Quote status:

DRAFT SENT ACCEPTED REJECTED EXPIRED

Accepted Quote can convert to Job.

24. SERVICE CATALOG / PRICE BOOK

Items:

- service;
- product;
- package.

Fields:

- name;
- description;
- unit;
- price;
- tax behavior;
- active status.

Historical Quotes/Invoices store snapshots.

Future Price Book updates must **never alter historical documents**.

25. JOB MANAGEMENT

Job is the central operational entity.

Fields include:

- job number;
- organization;
- customer;
- lead;
- quote;
- service;
- description;
- priority;
- status;
- assigned team;
- scheduled period;
- price estimate;
- tags;
- internal notes.

26. JOB STATUS

Default:

DRAFT

CONFIRMED

SCHEDULED

IN_PROGRESS

ON_HOLD

COMPLETED

CANCELLED

Industry template may simplify this.

27. JOB DETAIL PAGE

Header:

Job number
Customer
Status
Priority

Primary actions:

Schedule Assign Start Complete Invoice

Tabs:

Overview
Visits
Checklist
Photos
Materials
Notes
Activity
Finance

28. VISIT / APPOINTMENT

Visit fields:

- job_id;
- worker;
- date;
- start time;
- expected duration;
- location;
- status;
- arrival time;
- start time;
- completion time.

Visit statuses:

SCHEDULED EN_ROUTE ARRIVED IN_PROGRESS COMPLETED CANCELLED NO_SHOW

29. SCHEDULING ENGINE

Calendar views:

- Day
- Week
- Month
- Worker

Scheduling must consider:

- worker availability;
- leave;
- overlapping visits;
- duration;
- multiple workers;
- timezone.

Schedule conflict triggers warning.

Owner may override.

Override must record:

- actor;
- timestamp;
- reason if required.

30. WORKER MOBILE EXPERIENCE

Worker home:

Today

Cards:

09:00 — Service AC Customer Address Status

Actions:

Maps Call Depart

Job detail shows only necessary information.

31. WORKER FLOW

Worker:

1. Opens assigned Visit.
2. Tap **Depart**.
3. Tap **Arrived**.
4. Tap **Start Work**.
5. Completes checklist.
6. Uploads required photos.
7. Adds notes.
8. Tap **Finish**.

System updates:

- visit;
- job;
- dashboard;
- timeline;
- customer history;
- automation triggers.

32. OFFLINE-TOLERANT FIELD WORK

Worker app should tolerate unstable connectivity.

Actions that may queue locally:

- status updates;
- checklist;
- notes;
- photo uploads.

UI must show:

Pending sync

Synced

Failed — Retry

Never pretend an upload has succeeded when it has not.

33. PHOTOS

Support:

- before;
- during;
- after;
- documents.

Client compresses images before upload where appropriate.

Photo state:

UPLOADING

UPLOADED

FAILED

If required evidence is missing:

Worker receives clear warning before job completion.

34. CHECKLIST

Industry templates provide default checklists.

Examples AC:

- inspect filter;
- check temperature;
- clean unit;
- verify drainage;
- test operation.

Checklist supports:

- checkbox;
 - number;
 - text;
 - yes/no;
 - photo;
 - signature later.
-

35. CUSTOMER ASSETS

Asset examples:

AC
Vehicle
CCTV system
Router
Generator
Machine

Fields:

- asset type;
- customer;
- brand;
- model;
- serial number;
- installation date;
- warranty;
- notes;
- photos.

Asset has service history.

36. RECURRING SERVICE

Recurring schedules support:

- daily;
- weekly;
- monthly;
- yearly;
- custom interval.

System must **not create years of jobs upfront.**

Generate only near-term occurrences.

Example:

next 30–60 days.

Historical Jobs remain immutable.

Changing recurring rule only affects future occurrences.

37. INVOICES

Invoice statuses:

DRAFT

ISSUED

PARTIALLY_PAID

PAID

OVERDUE

VOID

Invoice includes:

- invoice number;
- customer snapshot;
- items;
- amount;
- discount;
- tax;
- total;
- due date;
- payment status.

38. MONEY DATA

Never use binary floating point for money.

Use:

- integer minor units; or
- exact decimal.

Example:

Rp150.000

must never become:

149999.999999.

39. INVOICE IMMUTABILITY

Once invoice is issued:

Line items are snapshoted.

Changes to:

- service price;
- customer address;
- tax settings;

must not silently modify historical invoices.

Major changes require:

- revision;
 - void/reissue;
 - audit event.
-

40. PAYMENT TRACKING

Initial methods:

- cash;
- bank transfer;
- QR payment;
- card;
- other.

Fields:

- invoice;
- amount;
- payment date;
- method;
- reference;
- received by;
- notes.

Multiple payments may belong to one invoice.

41. FOLLOW-UP

Main screen:

Follow-ups Today

Each record:

Customer
Reason
Due
Assigned Sales

Actions:

Chat Done Snooze

Follow-up reasons:

- lead;
 - quote;
 - post-job;
 - payment;
 - reactivation;
 - recurring service.
-

42. WHATSAPP SHORTCUT

Available to all plans.

Action:

Chat WhatsApp

Opens WhatsApp using normalized customer phone.

Does not require API.

43. WHATSAPP PLATFORM INTEGRATION

Pro and Ultra may connect WhatsApp Business Platform.

Primary first-release purpose:

Inbound WhatsApp → Customer/Lead

Flow:

Incoming WhatsApp event

↓

Validate webhook

↓

Check idempotency

↓

Normalize phone

↓

Find customer

↓

Existing customer?

No → Create Customer

→ Create Lead/Request

Yes → Link activity

→ Optionally create new Lead/Request according to rule

44. WHATSAPP HISTORICAL CONTACT RULE

Connecting WhatsApp does **not** automatically import every historical WhatsApp contact.

Example:

Owner has 1,000 WhatsApp contacts.

After connecting ARVENA:

ARVENA does not suddenly create 1,000 Customers.

Customers enter via:

- new supported inbound events;

- CSV import;
 - manual entry;
 - explicit supported migration tools later.
-

45. WHATSAPP OUTBOUND RULE

WhatsApp messaging is not unlimited.

ARVENA must support:

- credit;
- quota;
- usage-based billing;
- top-up.

Meta requires businesses to follow WhatsApp Business messaging policies, including opt-in expectations and approved templates for certain business-initiated messaging. Integration must therefore enforce compliant messaging rules rather than behaving like an unrestricted bulk sender.

46. WHATSAPP CONNECTION HEALTH

Integration screen:

Connected

or

Connection Problem

Show:

- connected number;
- last successful event;
- last error;
- reconnect action.

ARVENA core remains operational during WhatsApp outage.

47. AUTOMATION ENGINE

Automation follows:

Trigger → Conditions → Action

Initial triggers:

lead_created

quote_accepted

job_scheduled

visit_completed

job_completed

invoice_issued

invoice_overdue

customer_inactive

48. INITIAL AUTOMATION TEMPLATES

Post-job follow-up

Job completed
→ wait 3 days
→ create follow-up

Appointment reminder

Visit tomorrow
→ create/send reminder

Invoice reminder

Invoice overdue 3 days
→ reminder

Reactivation

No activity 180 days
→ add reactivation task

49. REMINDER SAFETY

Before scheduled communication executes:

system re-checks state.

Example:

Reminder created yesterday.

Visit cancelled today.

Tomorrow:

ARVENA must detect cancellation and **not send reminder**.

50. BACKGROUND JOB ARCHITECTURE

Background operations include:

- reminders;
- imports;
- webhook processing;
- automation;
- notification dispatch;
- media processing.

Use durable queues.

With Supabase implementation, Postgres-native Queues/ `pgmq` are appropriate for durable background processing, while Cron can initiate scheduled work.

Every external/background job requires:

- idempotency key;
 - retry;
 - status;
 - error log;
 - dead-letter/failure handling.
-

51. IDEMPOTENCY

Duplicate events must never create duplicate business records.

Examples:

WhatsApp webhook delivered twice.

User double-clicks Save.

Payment callback retries.

System uses:

```
external_event_id
```

or

```
idempotency_key
```

with database uniqueness constraints.

52. NOTES

Internal notes can belong to:

- customer;
- lead;
- job;
- visit;
- invoice.

Notes support:

- author;
 - timestamp;
 - visibility;
 - mention/team notification later.
-

53. ACTIVITY TIMELINE

Activity Timeline answers:

What operationally happened?

Examples:

10:12 Lead created

10:30 Sales contacted customer

11:15 Quote sent

14:00 Job created

Tomorrow 09:00 Worker assigned

54. AUDIT LOG

Audit log answers:

Who changed important system/business data?

Examples:

Invoice changed from 1.5M → 1.2M

Permission changed.

Worker deleted.

Payment modified.

Ultra receives full audit capabilities.

Activity Timeline and Audit Log are different systems.

55. TEAM MANAGEMENT

Fields:

- name;
- email;
- phone;

- role;
- active status;
- branch;
- schedule;
- skills;
- join date.

Actions:

Invite

Disable

Change role

Assign branch

Disabling user must not delete historical activity.

56. TEAM PERFORMANCE

Metrics:

- jobs assigned;
- jobs completed;
- completion rate;
- on-time arrival;
- average completion time;
- follow-up completion for sales.

Avoid simplistic ranking that encourages bad behavior.

Metrics should provide operational insight rather than punishment.

57. REPORTING

Core reports:

Sales

Lead source

Conversion

Won/lost

Operations

Jobs created
Jobs completed
Delayed jobs
Job duration

Finance

Invoices issued
Payments received
Outstanding
Overdue

Customers

New customers
Repeat customers
Inactive customers

Team

Job completion
Worker workload

58. SEARCH

Global search supports:

- customer name;
- phone;
- job number;
- invoice number;
- asset serial;
- lead.

Search results respect permissions and organization boundaries.

59. IMPORT

CSV/Excel import supports:

- customers;

- basic assets;
- service catalog.

Flow:

Upload
↓
Column mapping
↓
Preview
↓
Validate
↓
Duplicate detection
↓
Import

Before import show:

Valid: 822
Duplicates: 43
Errors: 17

User can fix before committing.

Import must be recoverable/traceable.

60. EXPORT

Basic data export is available even on Free.

Reason:

Customer data belongs to the customer/business.

ARVENA must not create artificial data hostage behavior.

Export examples:

- customers;
 - jobs;
 - invoices;
 - payments.
-

61. SETTINGS

Settings groups:

Business

Name
Logo
Address

Localization

Country
Currency
Timezone
Locale

Taxes

Tax settings

Team

Roles
Users

Operations

Job statuses
Visit statuses

Templates

Industry
Checklist

Integrations

WhatsApp
Maps
Payments later

Billing

Plan
Usage

62. UI/UX DESIGN PRINCIPLES

ARVENA UI follows:

Clear before clever

No complex decorative interface.

One dominant action per screen

Example Job Detail:

Complete Job

must be visually clearer than secondary actions.

Progressive disclosure

Advanced controls hidden until needed.

Recognition over memory

Users should not memorize statuses or commands.

Minimum repeat entry

Existing information reused automatically.

63. DESKTOP UI

Desktop layout:

Left sidebar.

Top contextual bar.

Main content.

Optional right detail drawer.

Use tables for:

- leads;
- customers;
- jobs;
- invoices.

Use cards only where they improve scanning.

64. MOBILE UI

Tables transform into cards.

Primary actions remain thumb-accessible.

No horizontal scrolling for core workflows.

Mobile owner must support:

- dashboard;
 - customers;
 - create job;
 - assign;
 - reschedule;
 - invoices;
 - follow-up.
-

65. FORM UX

Forms use sections.

Example Create Job:

Customer

Service

Schedule

Assignment

Details

Avoid displaying 30 fields simultaneously.

Optional advanced fields collapsed.

Autosave drafts where valuable.

66. SMART DEFAULTS

When creating Job from Customer:

Automatically populate:

- customer;
- phone;
- address;
- timezone.

When Job originates from Quote:

Automatically populate:

- service items;
 - price;
 - customer;
 - notes where relevant.
-

67. LOADING STATES

Do not show blank white screen.

Use skeletons.

Button submitting state:

Creating Job...

Disable duplicate submission.

68. EMPTY STATES

Bad:

No data.

Good:

Belum ada pekerjaan.

Buat pekerjaan pertama untuk mulai mengatur jadwal tim Anda.

Button:

Create Job

69. ERROR STATES

Never:

Error 500.

Instead:

Job belum berhasil disimpan.

Data Anda belum berubah. Periksa koneksi lalu coba lagi.

Actions:

Retry Save Draft

Error message should explain:

1. what failed;
2. what remains safe;
3. what the user can do.

70. CONFIRMATIONS

Confirmation required for:

- cancel job;
- void invoice;
- remove user;
- delete recurring schedule;
- disconnect integration.

Simple edits do not need confirmation.

Avoid confirmation fatigue.

71. ACCESSIBILITY

Target:

WCAG-conscious interface.

Requirements:

- sufficient contrast;
 - keyboard navigation;
 - visible focus states;
 - proper labels;
 - non-color-only status indicators;
 - scalable text.
-

72. DESIGN SYSTEM

Core components:

Button
Input
Textarea
Select
Combobox
Date Picker
Time Picker
Badge

Avatar
Table
Card
Modal
Drawer
Toast
Tabs
Dropdown
Skeleton
Empty State
Error State
Pagination
Command Search

Components must be reusable.

No page-specific visual hacks unless necessary.

73. STATUS COLOR SYSTEM

Colors must be semantic and consistent.

Examples:

Green: completed/paid/success

Yellow: attention/pending

Red: overdue/error/cancel risk

Blue: scheduled/information

Gray: draft/inactive

Never rely only on color.

Text status always shown.

74. RESPONSIVE BREAKPOINT BEHAVIOR

Desktop: full sidebar + table.

Tablet: collapsed sidebar + simplified columns.

Mobile: bottom nav + cards + drawers.

Features must not disappear merely because screen is smaller.

75. TECHNICAL REFERENCE ARCHITECTURE

Recommended baseline:

Frontend

React/Next.js-style web application

TypeScript

Responsive PWA

Backend/Data

PostgreSQL

Supabase platform where practical

Auth

Supabase Auth or equivalent

Data Security

Postgres Row-Level Security

Storage

Object storage

Background Processing

Durable queue + workers

Scheduling

Cron/scheduler

External Connectors

Server-side APIs/webhooks

76. DOMAIN ARCHITECTURE

Code organized around domains:

CRM

Work Management

Scheduling

Billing

Communication

Automation

Reporting

Templates

Identity & Access

Avoid organizing entire system only by UI pages.

77. CORE DATA ENTITIES

Primary entities:

organizations

organization_members

roles

permissions

customers

customer_addresses

leads

lead_status_history

quotes

quote_items

jobs

job_items

visits

visit_assignments

checklists

checklist_responses

assets

asset_service_history

invoices

invoice_items

payments

follow_ups

notes

activities

audit_logs

recurring_rules

automation_rules

automation_runs

notifications

integrations

external_events

media

service_catalog

branches

78. DATABASE REQUIREMENTS

Every tenant-scoped table must contain:

organization_id

Relevant indexes include:

organization_id

customer_id

assigned_worker_id

status

scheduled_at

created_at

Indexes must follow real query patterns.

79. ROW LEVEL SECURITY

RLS is mandatory on exposed tenant tables.

Supabase documentation explicitly recommends enabling RLS for tables exposed through its API.

Rules:

Owner: organization-wide access.

Worker: assigned records only.

Sales: permitted sales scope.

Finance: financial scope.

Automated tests must test both:

ALLOW cases

and

DENY cases

Security tests are release blockers.

80. SERVICE ROLE SECURITY

Service credentials:

- server-only;
 - never exposed to browser;
 - never bundled into client application;
 - secrets stored securely.
-

81. SECURITY REQUIREMENTS

Required:

- secure authentication;
 - tenant isolation;
 - least privilege;
 - audit logs;
 - input validation;
 - rate limiting;
 - secure webhook verification;
 - secret rotation capability;
 - secure file access;
 - signed/private media URLs where needed.
-

82. DATA PRIVACY

ARVENA should be designed toward compliance readiness for:

- Indonesia PDP requirements;
- GDPR-style data rights where applicable.

Capabilities:

- export;
- account deletion workflow;
- data retention;
- privacy settings;
- access control;
- auditability.

Legal implementation requires jurisdiction-specific review before global launch.

83. STORAGE SECURITY

Media path should logically include:

```
organization / customer / job / file
```

Access controlled through authenticated authorization.

Worker cannot browse unrelated organization media.

84. BACKUP STRATEGY

Database:

daily backup in production according to chosen infrastructure plan.

Storage:

separate backup process.

Supabase database backup does not automatically restore actual Storage API objects; only database metadata is included, so ARVENA must maintain a separate media backup/export strategy.

85. OBSERVABILITY

Production monitoring includes:

- application errors;
- API latency;
- failed queue jobs;
- failed webhooks;
- failed automations;
- database errors;
- storage failures;
- integration status.

Every error should have traceable ID.

Example:

ARV-WA-1042

86. FAILURE SCENARIOS

Duplicate customer

Cause: different phone formats.

Protection: phone normalization + unique organization scope.

Webhook delivered twice

Protection: idempotency.

Double Save

Protection: disable button + idempotency.

Worker schedule overlap

Protection: conflict warning.

Offline worker

Protection: pending local queue + sync.

Photo failure

Protection: retry and upload status.

Invoice simultaneous edit

Protection: optimistic concurrency/version number.

WhatsApp outage

Protection: connector isolation.

Automation failure

Protection: queue + retry + logs.

Reminder after cancellation

Protection: state re-check at execution.

Import corruption

Protection: preview + validation + import log.

87. CONCURRENCY CONTROL

Mutable critical records use version control.

Example:

Invoice version 4.

Two admins edit simultaneously.

Second save detects stale version.

UI says:

Invoice berubah sejak Anda membukanya.

Review versi terbaru sebelum menyimpan.

Never silently overwrite critical data.

88. SOFT DELETE

Default deletion behavior:

archive.

Entities such as:

- customer;
- worker;
- service item;
- job;

are archived when possible.

This preserves history.

89. NOTIFICATIONS

Channels:

Internal

In-app notifications.

Email

Operational/account notifications.

WhatsApp

Pro/Ultra where connected and permitted.

Notification preference configurable.

90. PRODUCT TIERS

ARVENA FREE

Purpose:

Rapikan usaha.

Price:

Rp0

Launch limits:

- 1 organization;
- owner + 2 additional team members;
- 30 jobs/month;
- customer database;
- manual leads;
- jobs;
- visits;
- calendar;
- basic assets;
- quotes;
- invoices;
- payment tracking;
- basic recurring;
- checklist;
- photo evidence;
- limited media storage;
- follow-up;
- WhatsApp shortcut;
- Maps shortcut;
- basic reports;
- import;
- export.

Customer count is not artificially capped at a tiny number.

91. ARVENA PRO

Purpose:

Kurangi kerja admin.

Price:

Rp300.000/month

Includes Free plus:

- larger team allowance;
- significantly higher job limits;
- expanded storage;
- WhatsApp connection;
- inbound customer/lead creation;
- automation templates;
- reminders;
- advanced recurring workflows;
- branded quote/invoice;
- Service Catalog;
- custom fields;
- team performance;
- richer reports;
- multiple admins;
- integration health.

WhatsApp message costs/credits are separate from unlimited subscription usage.

92. ARVENA ULTRA

Purpose:

Kontrol dan otomasi bisnis.

Price:

Rp799.000/month

Includes Pro plus:

- multi-branch;
- advanced automation;
- customer reactivation;
- advanced analytics;
- advanced assets/warranty;
- approval flows;
- custom roles;
- full audit log;
- recurring contracts;
- SLA/ticket capabilities;
- advanced reporting;
- advanced integrations/API access where applicable.

Usage-heavy resources may have fair-use/add-on limits.

93. ENTITLEMENT ENGINE

Plan limitations must not be hardcoded across frontend pages.

Use centralized entitlements.

Example:

```
feature.whatsapp_connection
```

```
feature.advanced_automation
```

```
limit.jobs_monthly
```

```
limit.team_members
```

```
limit.storage_bytes
```

This allows changes without rewriting product logic.

94. FEATURE GATING PRINCIPLE

Do not lock an industry's essential workflow behind Ultra.

Example:

If recurring scheduling is essential for cleaning businesses:

Basic recurring remains available on Free.

Pro adds automation and scale.

Ultra adds contract sophistication.

Difference between tiers:

scale + automation + integrations + management sophistication

not intentionally broken workflows.

95. USAGE PAGE

Users can see:

Jobs: 21 / 30

Storage: 340 MB / 500 MB

Team: 3 / 3

Messaging: credit balance

No surprise limits.

96. BUSINESS MODEL SAFETY

Avoid unlimited high-cost usage for:

- WhatsApp messages;
- AI;
- storage;
- heavy external APIs.

Add-on system should allow:

- messaging credits;
- storage;
- team seats;

- additional branches;
 - usage packs.
-

97. OUT OF SCOPE — V1

ARVENA v1 will **not** attempt to become:

- full accounting ERP;
- payroll platform;
- banking product;
- marketplace;
- 24-hour employee GPS tracker;
- complex route optimization platform;
- procurement ERP;
- manufacturing ERP;
- financing platform;
- full AI receptionist;
- Zapier clone;
- full omnichannel call center.

These can be evaluated later.

98. PRODUCT ANALYTICS

Track anonymous/product usage events such as:

`onboarding_completed`

`customer_created`

`job_created`

`job_completed`

`invoice_issued`

`invoice_paid`

`followup_completed`

`automation_enabled`

whatsapp_connected

Metrics must respect privacy requirements.

99. NORTH STAR METRIC

Primary product success signal:

Organizations completing real operational jobs through ARVENA every week.

Not merely registered accounts.

100. ACTIVATION METRIC

Organization activated when it has:

1. created/imported customer;
2. created first job;
3. scheduled/assigned work;
4. completed operational action.

Target:

first activation within **10 minutes** where possible.

101. RETENTION METRIC

Critical question:

After 30 days, does the owner continue running their business through ARVENA without being chased?

Key signals:

- weekly active organization;
 - jobs completed;
 - invoices issued;
 - recurring jobs;
 - follow-ups performed.
-

102. SALES VALIDATION

Initial market validation should target real owners.

Early target:

50–100 prospects.

10–20 active trials.

1–5 paid customers within initial validation period is a positive signal.

Feedback should focus on:

- confusion;
 - missing critical workflow;
 - duplicated admin work;
 - willingness to pay;
 - retention.
-

103. PERFORMANCE TARGETS

Typical page interaction should feel near-instant.

Targets:

- primary navigation responsive;
- optimistic UI where safe;
- lists paginated;
- indexes for common filters;
- images lazy-loaded;
- background operations not block UI.

Large reports may process asynchronously.

104. LIST PAGINATION

Large organizations must not load all rows.

Use pagination/cursor patterns.

Filters and search executed server-side/database-side where appropriate.

105. TIMEZONE MODEL

Store canonical timestamps in UTC.

Render according to organization/user timezone.

Recurring rules operate with explicit timezone context.

Prevent DST problems in global markets.

106. ADDRESS MODEL

Address supports:

- label;
- line1;
- line2;
- city;
- administrative region;
- postal code;
- country;
- latitude;
- longitude.

Do not assume Indonesian administrative hierarchy.

107. MAPS

Maps provider accessed through abstraction.

Initial UX may use:

Open in Maps

Later:

embedded maps/routing.

Provider must be replaceable.

108. CUSTOM FIELDS

Pro+ supports custom fields.

Entity targets:

- Customer
- Lead
- Job
- Asset

Types:

text
number
date
boolean
select

Custom fields are metadata-driven.

No new database column for every customer custom field.

109. INDUSTRY TERMINOLOGY

Terminology config examples:

Generic: Job

Workshop: Service Order

IT: Ticket

Event: Project

Backend entity remains stable.

UI label changes through configuration.

110. TEMPLATE ENGINE

Industry template contains:

- terminology;
- lead stages;
- job statuses;
- checklist;
- custom fields;
- recommended modules;
- service catalog suggestions.

Templates can be updated without forking codebase.

111. CUSTOM BUSINESS MODE

If industry is not listed:

Select:

Other / Custom Business

ARVENA starts with generic workflow.

User customizes:

- terminology;
- statuses;
- fields;
- modules.

This makes multi-industry expansion possible.

112. APPROVAL SYSTEM — ULTRA

Approval examples:

- discount > threshold;
- invoice void;
- expense later;
- job cancellation;
- quote amount.

Approval requests record:

requester
approver
status
timestamp
decision note

113. MULTI-BRANCH — ULTRA

Branch contains:

- name;
- address;
- timezone;
- team;
- jobs;
- customers where applicable.

Owner can view:

All Branches

or filter one branch.

Permissions may be branch-scoped.

114. SLA / TICKETING — ULTRA MODULE

For service contracts:

- ticket priority;
- response SLA;
- resolution SLA;
- escalation;
- contract reference.

Not displayed to businesses that do not enable the module.

115. BUSINESS HEALTH CENTER

This is a strategic differentiator.

Instead of only saying:

Revenue = X

ARVENA says:

6 jobs belum memiliki worker.

9 leads belum di-follow-up >24 jam.

Rp12.300.000 invoice overdue.

14 customers due for maintenance.

Each insight has CTA.

Example:

Review 9 Leads

ARVENA should help owner decide, not merely display data.

116. UX SUCCESS CRITERIA

A new owner should be able to:

Create customer: < 60 seconds.

Create job: < 90 seconds.

Assign worker: < 20 seconds.

Reschedule: < 30 seconds.

Worker start job: < 10 seconds.

Complete follow-up: < 15 seconds.

No training manual should be required for core flow.

117. CRITICAL END-TO-END ACCEPTANCE TEST

Scenario:

Customer requests AC service.

1. Admin creates Customer.
2. Admin creates Lead.
3. Lead converted to Job.
4. Job scheduled tomorrow 10:00.
5. Worker assigned.
6. Worker sees job on phone.
7. Worker taps Depart.
8. Worker opens Maps.
9. Worker arrives.
10. Worker starts work.
11. Worker completes checklist.
12. Worker uploads before/after photos.
13. Worker completes Visit.
14. Job becomes Completed according to rule.
15. Admin creates invoice.
16. Payment recorded.
17. Invoice becomes Paid.
18. System schedules post-service follow-up.
19. Customer history shows full lifecycle.
20. Owner dashboard updates correctly.

All must work without duplicate manual entry.

118. MULTI-VISIT ACCEPTANCE TEST

Job:

CCTV Installation.

Visits:

Survey
Installation
Testing

Completing Survey must not mark the entire Job Completed.

Job completes only according to job completion rule.

119. SCHEDULE CONFLICT TEST

Worker already booked:

10:00–12:00.

Admin tries:

11:00–13:00.

System must warn:

Worker has another appointment during this time.

Admin may override with explicit action.

120. DUPLICATE PHONE TEST

Customer entered as:

081912345678

Later:

+6281912345678

System must detect likely same customer within organization.

121. DUPLICATE WEBHOOK TEST

Same WhatsApp event delivered 3 times.

Expected:

1 external event.

At most 1 customer creation.

At most 1 configured lead creation.

122. OFFLINE TEST

Worker loses internet.

Worker:

- starts visit;
- adds checklist;
- takes photos;
- finishes work.

System clearly labels pending sync.

When connection returns:

actions synchronize safely.

No duplicate activity.

123. INVOICE INTEGRITY TEST

Invoice issued for:

Rp1.000.000.

Service Catalog later changes price to:

Rp1.200.000.

Historical invoice remains:

Rp1.000.000.

124. RECURRING TEST

Monthly AC service.

Rule changed after six months.

Completed jobs from previous months remain unchanged.

Only future occurrences adopt new configuration.

125. SECURITY TEST

Worker A in Organization A attempts API access to Customer from Organization B.

Expected:

DENIED

Even if worker knows record ID.

This is a release-blocking test.

126. ROLE TEST

Finance user attempts to change organization integration settings.

Expected:

Denied.

Worker attempts to view company-wide revenue.

Expected:

Denied.

Owner:

Allowed.

127. STORAGE TEST

Worker attempts to access photo belonging to another organization.

Expected:

Denied.

128. FAILURE RECOVERY TEST

Automation provider/API fails.

Expected:

- Job remains usable;
 - message queued/retried;
 - error visible;
 - no duplicate action;
 - admin can retry where appropriate.
-

129. DATA EXPORT TEST

Free organization exports customers.

Expected:

valid export.

No upgrade required simply to retrieve basic business data.

130. RESPONSIVE TEST

Core workflows tested at:

- desktop;
- tablet;
- common mobile widths.

No inaccessible buttons.

No core table requiring unusable horizontal navigation.

131. BROWSER / PWA TARGET

Initial launch:

Modern Chrome
Edge
Safari
Mobile Chrome
Mobile Safari

Native iOS/Android app is not required initially.

Responsive/PWA first.

132. IMPLEMENTATION PHASES

P0 — ARVENA CORE

Must be genuinely usable.

Includes:

- auth;
- organization;
- onboarding;
- roles;
- dashboard;
- customers;
- leads;
- quotes;
- jobs;
- visits;
- schedule;
- worker mobile;
- checklists;
- photos;
- notes;
- assets basic;
- invoices;
- payments;
- follow-up;
- recurring basic;
- reports basic;
- import/export;

- settings;
- responsive UI;
- loading/error/empty states;
- full acceptance tests.

P0 should already be sellable as Free/Core product.

133. P1 — ARVENA PRO

Includes:

- WhatsApp connector;
 - auto customer/lead intake;
 - automation templates;
 - advanced recurring;
 - Service Catalog;
 - custom fields;
 - branded documents;
 - team performance;
 - richer reports;
 - notification enhancements;
 - integration health;
 - usage/billing controls.
-

134. P2 — ARVENA ULTRA

Includes:

- multi-branch;
 - custom roles;
 - approval flows;
 - full audit;
 - advanced analytics;
 - advanced automation;
 - reactivation;
 - SLA/ticket module;
 - advanced assets;
 - recurring contracts;
 - external API/webhooks;
 - enterprise controls.
-

135. RELEASE GATE — P0

P0 cannot be called finished if any primary workflow still uses:

- PlaceholderPage;
- static dummy data;
- broken forms;
- unusable mobile layout;
- incomplete permissions;
- missing empty/error states.

Primary routes must be usable with real database data.

136. ENGINEERING QUALITY GATE

Before production release:

- lint passes;
 - typecheck passes;
 - unit tests pass;
 - integration tests pass;
 - RLS tests pass;
 - end-to-end tests pass;
 - production build passes;
 - migrations reviewed;
 - backup tested;
 - restore procedure documented;
 - security secrets reviewed;
 - performance smoke test completed.
-

137. PRODUCT QUALITY GATE

Before declaring ARVENA ready:

A test business must be able to run one full business day through ARVENA.

From:

customer request

to:

job completed

to:

invoice/payment

to:

follow-up

without needing spreadsheets for the core process.

138. UI QUALITY GATE

Every primary screen requires:

- desktop state;
- mobile state;
- loading;
- empty;
- error;
- populated;
- permission denied where relevant.

No screen should look unfinished.

139. UX QUALITY GATE

During usability test:

New user receives only the scenario.

They should successfully:

- add customer;
- create job;
- schedule worker;
- complete job;
- invoice customer.

Without step-by-step coaching.

Confusion points are treated as product defects.

140. FUTURE ROADMAP

Possible future capabilities:

- customer portal;
- route optimization;
- inventory purchasing;
- advanced subscription billing;
- customer self-booking;
- payment gateway integrations;
- AI operational assistant;
- AI call/chat assistant;
- advanced forecasting;
- marketplace;
- partner ecosystem;
- native apps.

They must not distract from core execution.

141. AI PRINCIPLE

AI is an enhancement.

Not core dependency.

ARVENA must still function without AI.

Potential AI future uses:

- summarize customer history;
- recommend follow-ups;
- categorize inbound requests;
- suggest schedule;
- detect operational risks;
- draft replies.

AI usage should be usage-based/add-on where costs are variable.

142. PRODUCT MOAT

ARVENA's defensibility should come from:

- excellent workflows;
- industry templates;
- historical customer/job data;
- recurring service intelligence;
- onboarding simplicity;
- operational automation;
- integrations;
- user experience;
- distribution network.

Not from hiding customer data.

143. FINAL PRODUCT EXPERIENCE

ARVENA should feel like:

“Saya tidak perlu lagi buka lima tempat untuk tahu bisnis saya sedang bagaimana.”

Owner opens ARVENA.

Sees what needs attention.

Admin knows what to schedule.

Sales knows who to follow up.

Worker knows where to go.

Finance knows who has not paid.

Customer history stays connected.

144. FINAL SYSTEM PHILOSOPHY

ARVENA is built around one concept:

Everything Connected.

Customer connects to Lead.

Lead connects to Quote.

Quote connects to Job.

Job connects to Visits.

Visits connect to Workers.

Job connects to Invoice.

Invoice connects to Payment.

Completion connects to Follow-up.

Customer connects to Assets.

Assets connect to future service.

All of that connects back to the Owner.

That is ARVENA.

145. LOCKED V1.0 PRODUCT DECISIONS

The following are considered locked baseline decisions:

Brand ARVENA.

Logo Official botanical A/root/growth logo approved before this PRD.

Positioning Service Business Operating System.

Architecture Universal Core + Industry Templates + Optional Modules.

Critical domain rule Job $\neq$ Visit.

Core philosophy One Input $\rightarrow$ Many Outcomes.

Dashboard Exception-driven.

Field UX Mobile-first.

Global model Global-ready from database design.

WhatsApp Connector, not core dependency.

Free strategy Useful product; no tiny arbitrary customer cap.

Paid strategy Scale + automation + integrations + management features.

Pricing Free / Rp300k Pro / Rp799k Ultra at launch baseline.

Technical principle Multi-tenant, RLS, idempotent, queue-driven background work.

Data philosophy Basic export available; business data is not held hostage.

Implementation P0 Core → P1 Pro → P2 Ultra.

Any future material change to these decisions must be made as a versioned product decision, for example:

ARVENA PRD v1.1

rather than silently changing the v1.0 baseline.

FINAL STATEMENT

ARVENA v1.0 is not intended to win by having the largest number of features.

It is intended to win because a service-business owner can open one system and clearly understand:

who the customers are,

what work must be done,

who is doing it,

when it happens,

whether it is completed,

whether it has been paid,

and

what should happen next.

If ARVENA achieves this with less friction than WhatsApp + spreadsheet + manual admin work, it has a foundation capable of serving many industries and expanding internationally.

ARVENA — Everything connected. Built to grow.